Recent advancements in AI/ML bring an unprecedented opportunity to foster inclusive technology design. While people with disabilities are often early adopters of technology [1], accessibility has historically been an afterthought in its development [2]. I am a **master's student in robotics at Carnegie Mellon University (CMU)** with **Professors Jeffrey Bigham** and **Aaron Steinfeld**. As a technical HCI researcher trained in computer science, I build novel interactive systems that use (and advance) ML to solve hard accessibility problems. With three years of research in enhancing information access, I am uniquely equipped to change this narrative by pushing for greater inclusion.

I view accessible computing as a critical lens for deepening our understanding of human-AI interaction. Through a Ph.D. from the School of Computer Science, I want to contribute to the technical foundations for AI systems that are more adaptive, responsible, and equitable. I aim to address some of these long-standing challenges through my future research: How can we build tools that dynamically respond to a long tail of user needs? How do we ensure our systems augment ethical user behaviors? How can technology cater to users from all backgrounds?

Below, I detail the research experiences that have informed my interests.

As a pre-doctoral researcher at Microsoft Research, I worked with **Dr. Saikat Guha** on making long-form print documents accessible to screen reader users. I initiated engagement with 32 visually impaired participants to establish a feedback loop, which iteratively unveiled three critical needs. First, they requested support for newspapers in their local languages, which led me to translate screen reader-accessible English e-newspapers. Second, they observed that the content of these newspapers often lacked local news, which was usually solely printed in local-language newspapers. I built a computer vision pipeline for converting print newspapers to screen reader-friendly formats (e.g., ePub) and persistently delivered them via a mailing list. The pipeline involved segmenting articles from newspaper pages and extracting their content using optical character recognition (OCR). Finally, they highlighted the problem of 'glanceability' in accessing these newspapers, as the linear narration by screen readers made it challenging to navigate or grasp the structure of the content quickly. To support glanceability, I undertook an empirical study to identify key visual cues for enhancing the aural experience. This work underscored the need for advanced OCR post-processing to retain document structure and styling and was presented at the **AVA Workshop at CVPR'22** [3]. This endeavor taught me how traditional AI systems often required optimization when accounting for diverse user needs.

I found the community's demand for access to print newspapers intriguing. Consequently, I examined the news consumption behaviors of blind and visually impaired (BVI) individuals with **Dr. Mohit Jain**. I conducted semi-structured interviews and contextual inquiries with 12 participants recruited through community-driven platforms. The study disclosed numerous operational barriers, such as the overwhelming presence of ads, paywalls, and popups on digital news platforms. Initial reviews of this work questioned the generalizability of our findings as our participants were skewed toward urban individuals. Recruiting participants from rural areas was particularly challenging, but through relentless outreach and snowball sampling, I successfully enlisted 5 additional participants to enhance diversity, analyzed the data thematically, and presented this work at **ASSETS'24** [4].

These projects demonstrated substantial efforts to retrofit accessibility into user interfaces (UI). In practice, it is a persistent challenge to ensure developers write the UI code that supports assistive technologies from the outset.

I joined **CMU's School of Computer Science** when AI coding assistants (e.g., GitHub Copilot) began transforming developer workflows. While they could potentially automate the production of accessible UI code, their impact remained unclear. Through a controlled task-based study with 16 developers, I found three barriers to realizing this opportunity: i) developers failed to prompt for accessibility considerations explicitly, ii) uncritically accepted incomplete code suggestions, and iii) struggled to detect potential accessibility issues in their code. I identified concrete feature requirements for tackling these behaviors: i) integrating system prompts for accessibility awareness, ii) generating code completion reminders, and iii) suggesting fixes for existing accessibility errors. I implemented these features using a multi-agent LLM architecture and consulted with GitHub Next to deploy CodeA11y, a standalone Copilot Extension plugin. I solved two complex technical problems in building CodeA11y: detecting real-time errors by integrating an accessibility linter and extracting relevant code context by reverse-engineering GitHub Copilot. A technical evaluation of CodeA11y with 20 developers demonstrated a statistically significant improvement in producing accessible UI code without compromising their satisfaction and ease of use. I presented the formative study as a poster at **ASSETS'24** [5], and the subsequent paper received positive reviews under **R&R for CHI'25**.

Besides my research, my involvement in mentoring and service has played a vital role in broadening my perspectives. I mentored a CMU undergraduate student in studying the risks of blind users relying on multimodal models like GPT-4v for visual assistance, revealing that hallucinated answers can be difficult for users to verify. This study was presented at the **VizWiz Grand Workshop at CVPR'24** [6]. Additionally, I organize the Accessibility Lunch weekly seminar at CMU, inviting researchers and accessibility advocates to learn from their work.

My research experiences in academia and industry have solidified my commitment to becoming a researcher. CMU RI and HCII are excellent fits for my interests, with many renowned faculty pursuing related research. I am excited about building upon my ongoing research on applying AI/ML for accessibility with my advisors, **Prof. Jeffrey Bigham** and **Prof. Aaron Steinfeld**. Additionally, I aspire to be involved in **Prof. <>**'s recent work on <>. I am also interested in collaborating with **Prof. <>** on <> and **Prof. <>** on <>. After spending the last year at CMU, I am confident that I belong in its research community, making it the ideal place for me to pursue my Ph.D.

References:

[1] Bigham, J. P., & Carrington, P. (2018). Learning from the front: People with disabilities as early adopters of AI. *Proceedings of the 2018 HCIC Human-Computer Interaction Consortium*.

[2] Churchill, E. F. (2018). Putting accessibility first. *Interactions*, *25*(5), 24-25.

[3] Mowar, P., Ganu, T., & Guha, S. (2022). Towards Optimizing OCR for Accessibility. *AVA: Accessibility, Vision, and Autonomy Meet (CVPR 2022 Workshop)*.

[4] Mowar, P., Gupta, M., & Jain, M. (2024, October). Breaking the News Barrier: Towards Understanding News Consumption Practices among BVI Individuals in India. In *Proceedings of the 26th International ACM SIGACCESS Conference on Computers and Accessibility* (pp. 1-11).

[5] Mowar, P., Peng, Y. H., Steinfeld, A., & Bigham, J. P. (2024, October). Tab to Autocomplete: The Effects of AI Coding Assistants on Web Accessibility. In *Proceedings of the 26th International ACM SIGACCESS Conference on Computers and Accessibility* (pp. 1-6).

[6] Agarwal, Y., & Mowar, P. (2024). Shifted reality: Navigating altered visual inputs with multimodal LLMs. *VizWiz Grand Challenge (CVPR 2024 Workshop)*.